

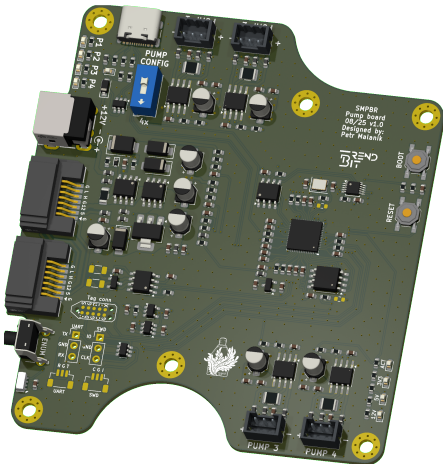
SMPBR / Pump_board

Rev 1.0

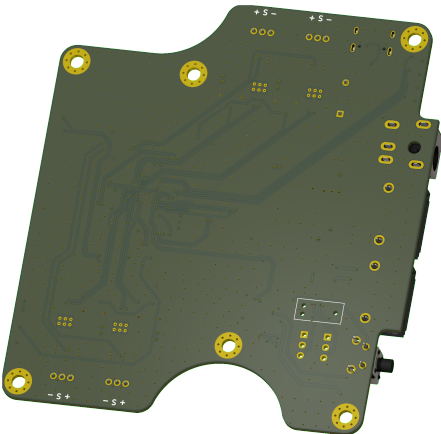
2025-09-23

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TOP VIEW



BOTTOM VIEW



NOTES

Not fitted components are marked as X

DESIGN CONSIDERATIONS

INFO NOTE:
Example text for informational design notes.

DESIGN NOTE:
Example text for critical design notes.

LAYOUT NOTE:
Example text for critical layout notes.

Designed for: TrendBit
Designed by: Petr_Malaník

Sheet: / File: pump_board.kicad_sch			Project: SMPBR	
Title:			Board: Pump_board	
Size: A4	Date: 2024-12-03	GIT hash: 48ae82b	Rev: 1.0	
KiCad E.D.A. 9.0.1		Id: 1/13		

Revision History



Designed for: TrendBit
Designed by: Petr_Malaník

Sheet: /Revision History/
File: revision_history.kicad_sch

Project: SMPBR

Title: Revision History

Board: Pump_board

Size: A4

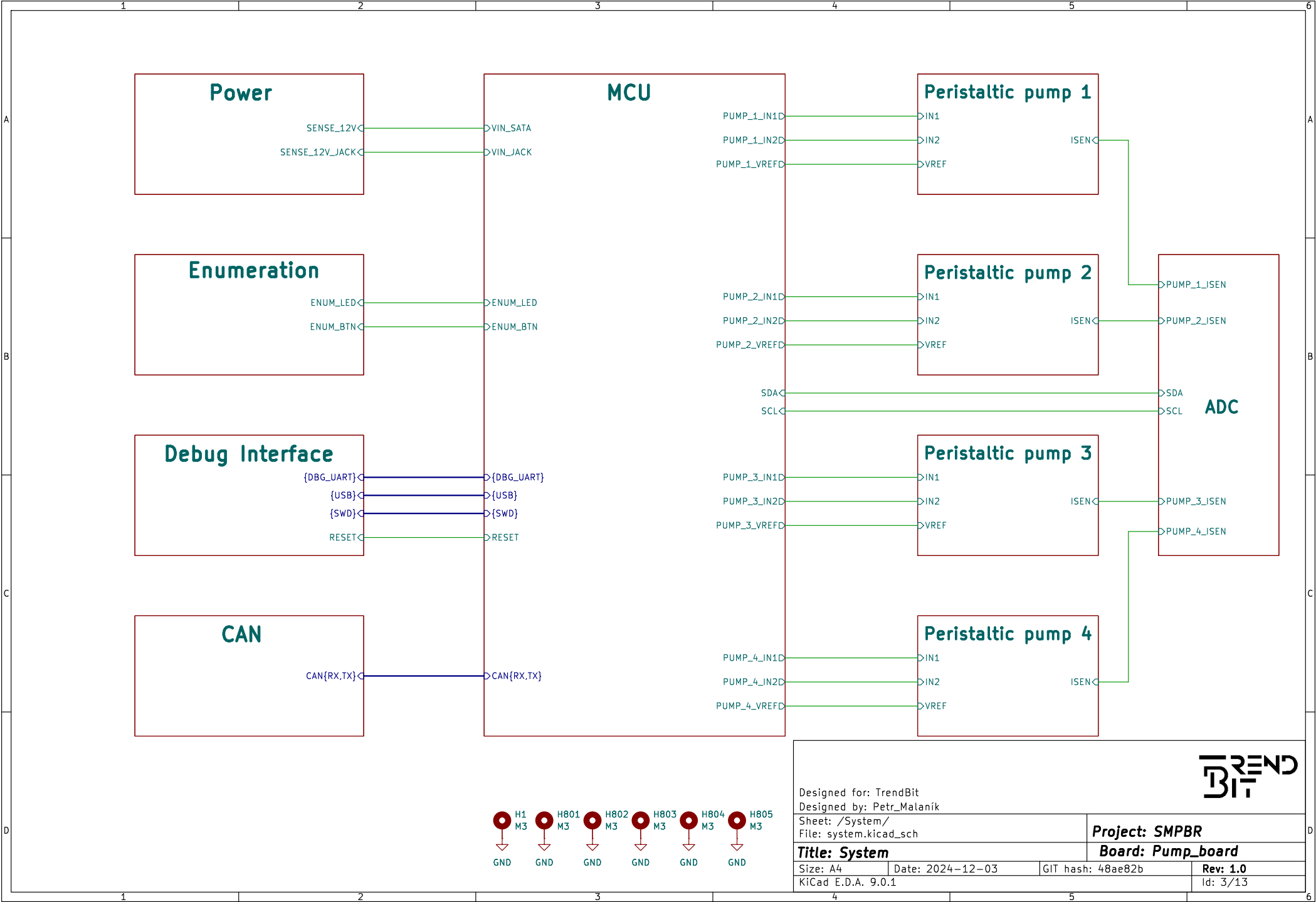
Date: 2024-12-03

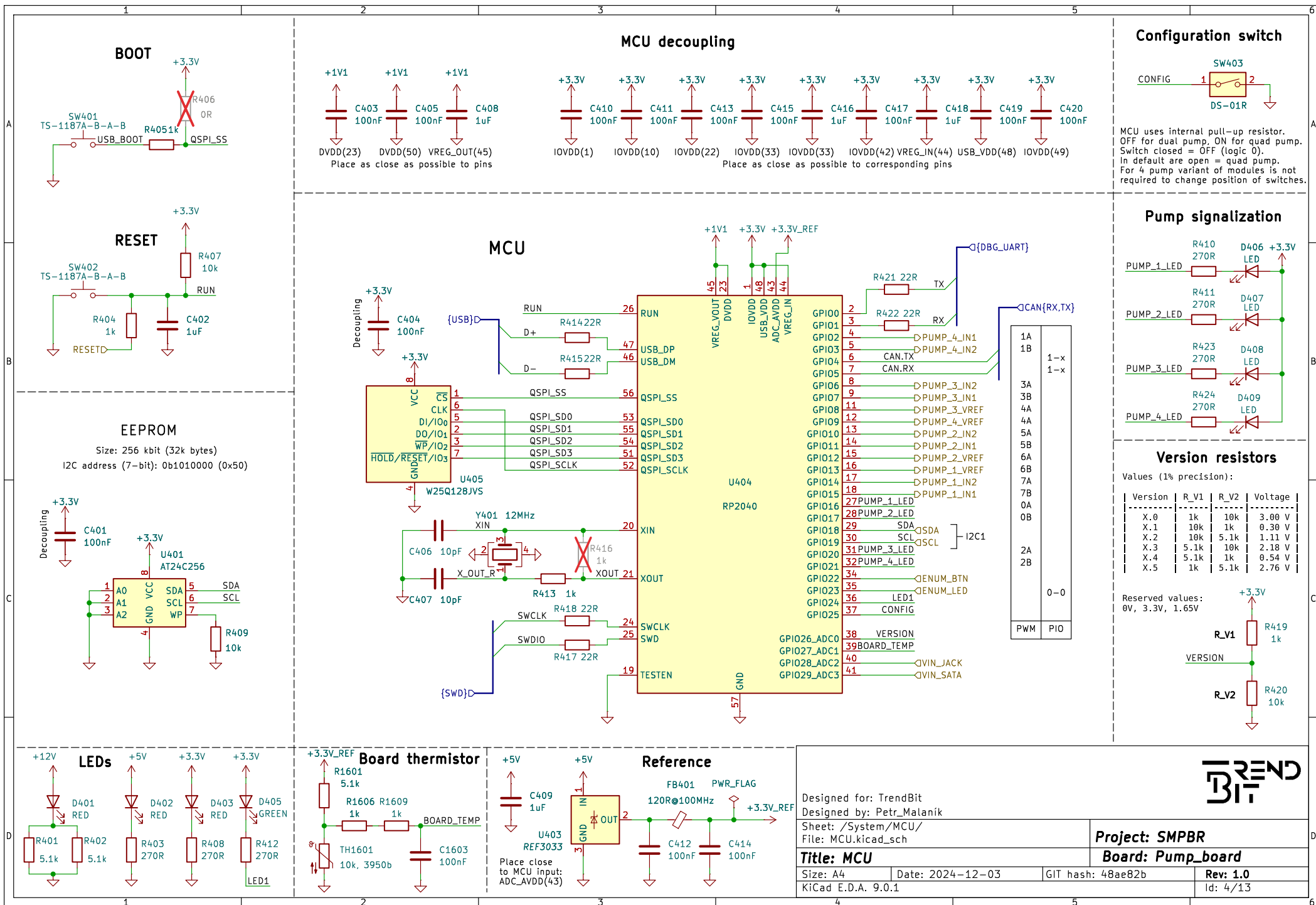
GIT hash: 48ae82b

Rev: 1.0

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Id: 2/13





TREND
BIT

Designed for: TrendBit
Designed by: Petr Malanik

Sheet: /System/MCU/
File: MCU.kicad_sch

Title: MCU

Size: A4

Date: 2024-12-03

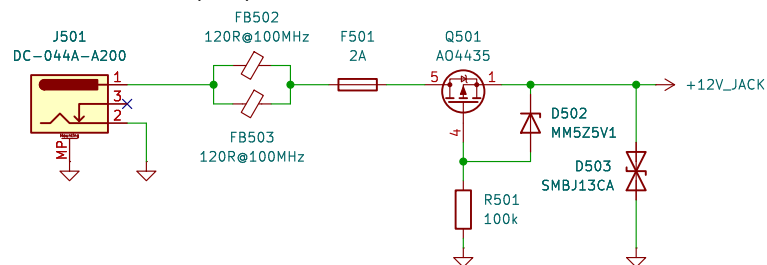
GIT hash: 48ae82b

Rev: 1.0
Id: 4/13

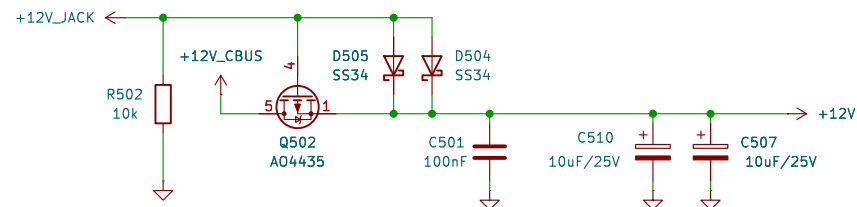
Project: SMPBR

Board: Pump board

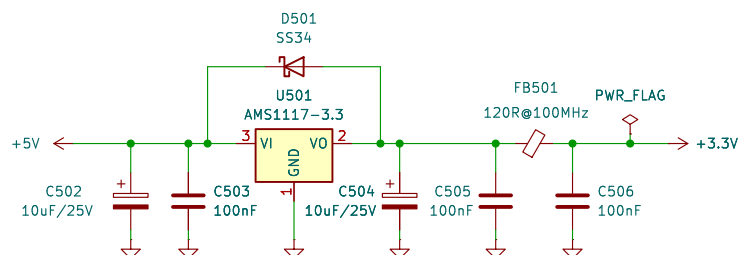
Input protection



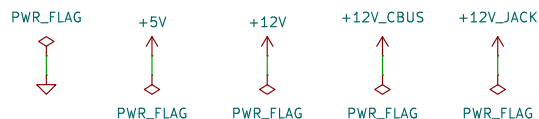
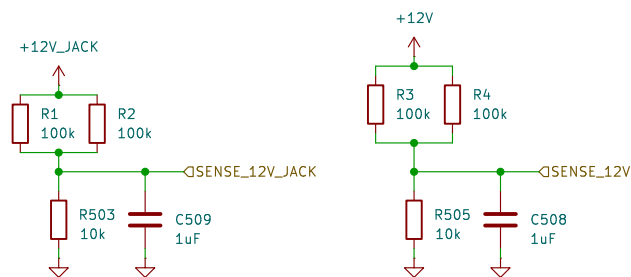
Supply switching



Supply - 3.3V



12V reading



Designed for: TrendBit
Designed by: Petr_Malanik

Sheet: /System/Power/
File: power.kicad_sch

Title: Power

Size: A4

Date: 2024-12-03

GIT hash: 48ae82b

Rev: 1.0

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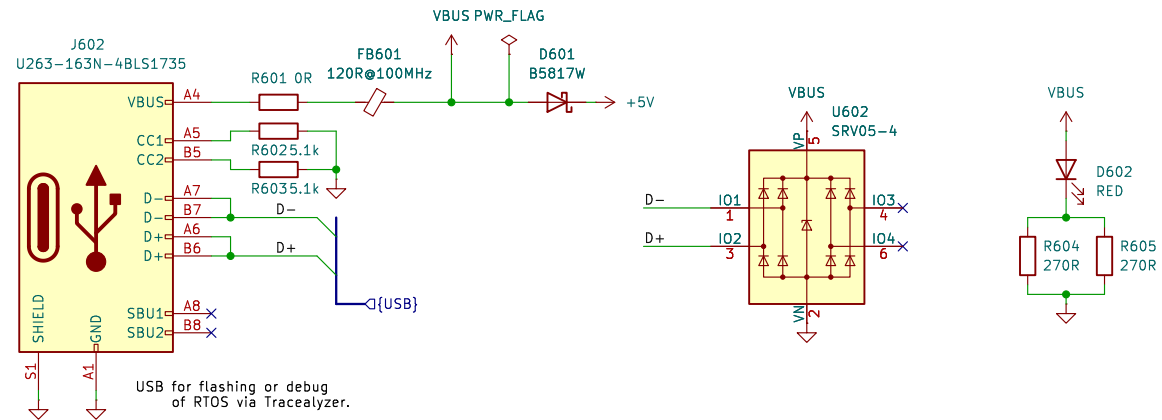
Id: 5/13



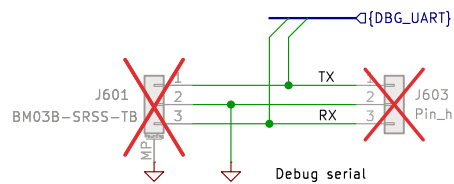
Project: SMPBR

Board: Pump_board

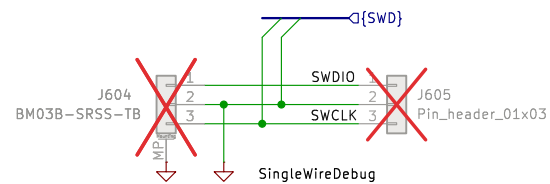
USB – Update & Debug



Serial Debug

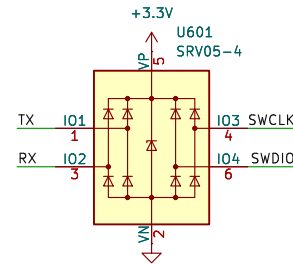


SWD Debug

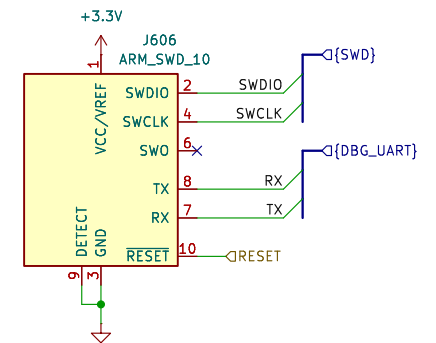


Pinout of these connectors is intended to be used with RPi debug Probe

DBG ESD Protection



Tag-connector



Designed for: TrendBit
Designed by: Petr_Malaník

Sheet: /System/Debug Interface/
File: debug.kicad_sch

Title: *Debug Interface*

Size: A4

Date: 2024-12-03

GIT hash: 48ae82b

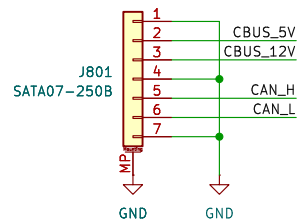
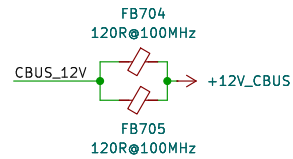
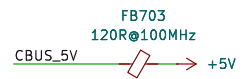
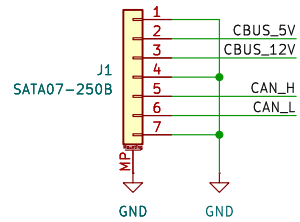
Rev: 1.0

SIZE: A1	5
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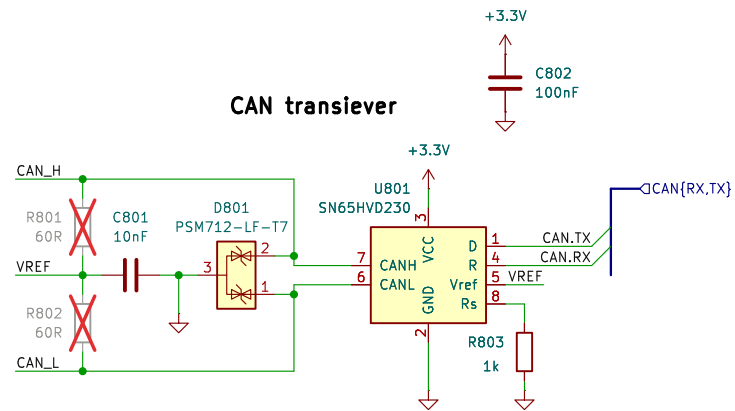
TREND
BIT

Project: SMPBR

Board:	Pump_board
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CAN transiever



Designed for: TrendBit
Designed by: Petr_Malaník

Sheet: /System/CAN/
File: can_interface.kicad_sch

Title: CAN

Project: SMPBR

Board: Pump_board

Size: A4

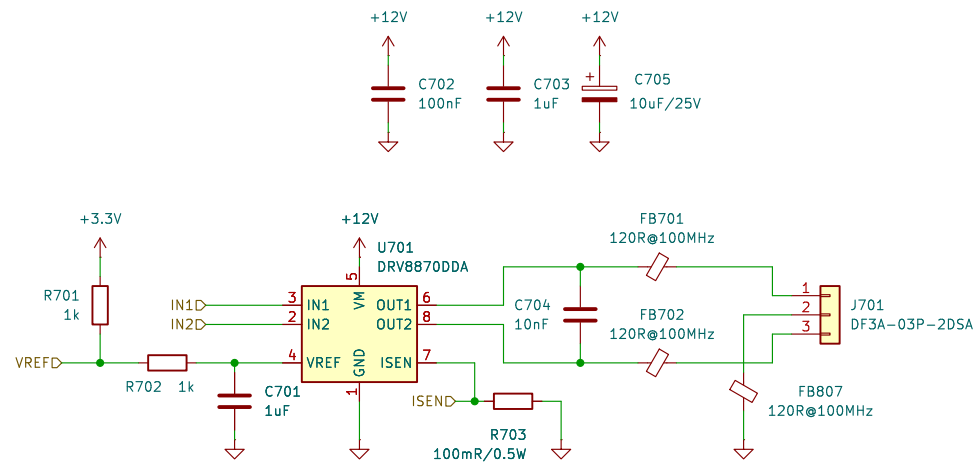
Date: 2024-12-03

GIT hash: 48ae82b

Rev: 1.0

KiCad E.D.A. 9.0.1

Id: 7/13



Input filter:
CTRL PWM frequency = 10 kHz
Settling time (10->90%) = ~4 ms
Peak to peak (Duty = 50%) = 0.07 V
Default max current = 3.3 A (with 0.1R Sense resistor)

$$R_{SEN} = V_{REF} / (10 \times I_{TRIP})$$

Current limit = 3.30 A



Designed for: TrendBit
Designed by: Petr.Malaník

Sheet: /System/Peristaltic pump 1/
File: motor.kicad_sch

Title: Peristaltic pump 1

Project: SMPBR

Board: Pump_board

Size: A4

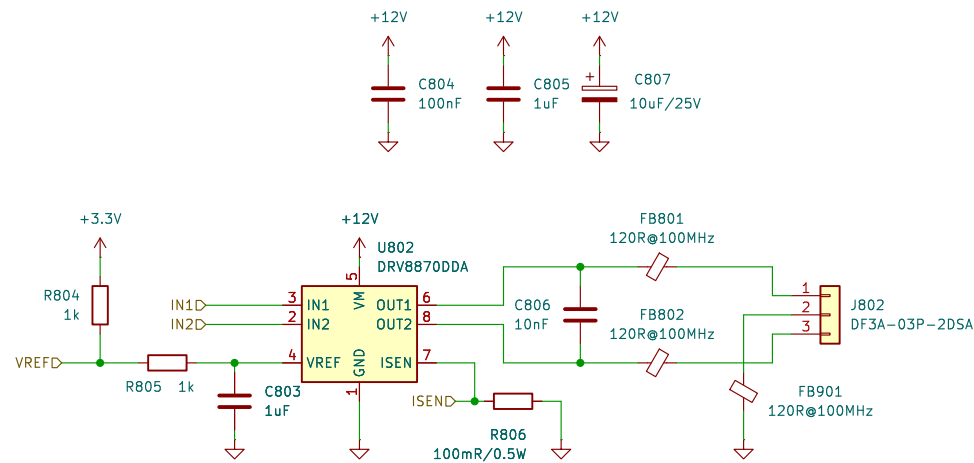
Date: 2024-12-03

GIT hash: 48ae82b

Rev: 1.0

KiCad E.D.A. 9.0.1

Id: 8/13



Input filter:
CTRL PWM frequency = 10 kHz
Settling time (10->90%) = ~4 ms
Peak to peak (Duty = 50%) = 0.07 V
Default max current = 3.3 A (with 0.1R Sense resistor)

$$R_{SEN} = V_{REF} / (10 \times I_{TRIP})$$

Current limit = 3.30 A



Designed for: TrendBit
Designed by: Petr.Malaník

Sheet: /System/Peristaltic pump 2/
File: motor.kicad_sch

Title: Peristaltic pump 2

Project: SMPBR

Board: Pump_board

Size: A4

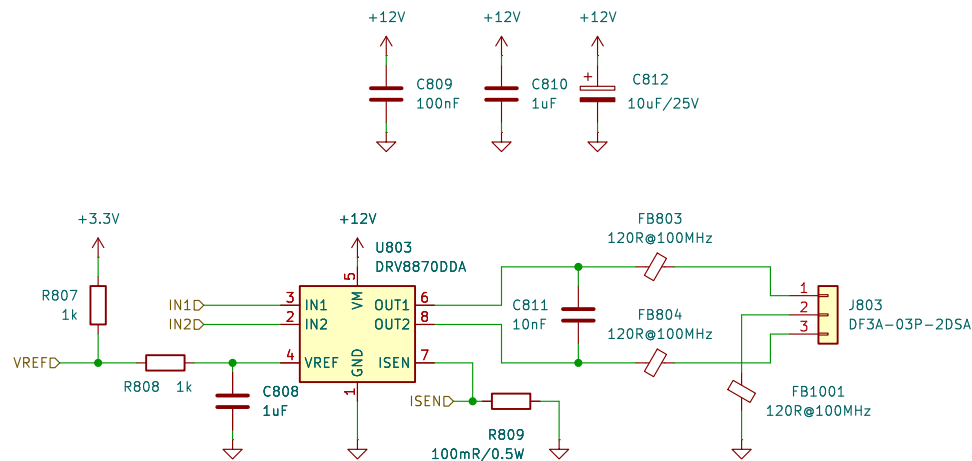
Date: 2024-12-03

GIT hash: 48ae82b

Rev: 1.0

KiCad E.D.A. 9.0.1

Id: 9/13



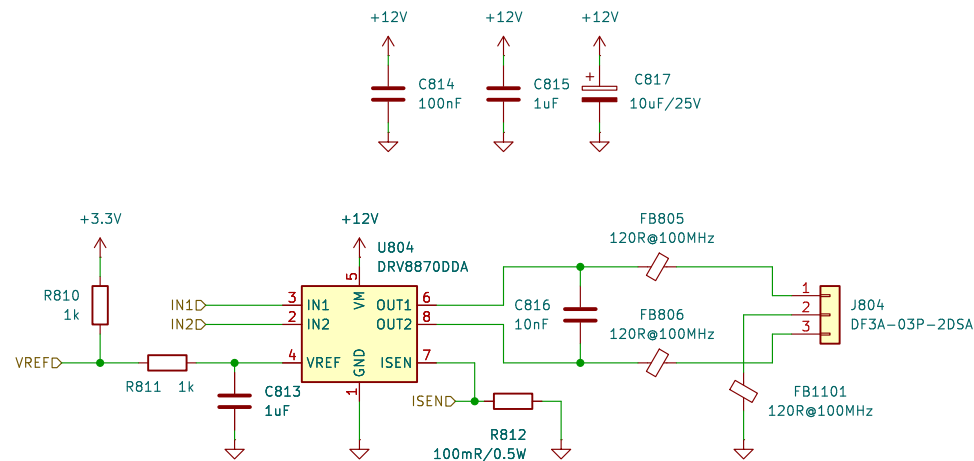
Input filter:
CTRL PWM frequency = 10 kHz
Settling time (10->90%) = ~4 ms
Peak to peak (Duty = 50%) = 0.07 V
Default max current = 3.3 A (with 0.1R Sense resistor)

$$R_{SEN} = V_{REF} / (10 \times I_{TRIP})$$

Current limit = 3.30 A



Designed for: TrendBit Designed by: Petr_Malanik			SB	
Sheet: /System/Peristaltic pump 3/ File: motor.kicad_sch			Project: SMPBR	
Title: Peristaltic pump 3			Board: Pump_board	
Size: A4	Date: 2024-12-03	GIT hash: 48ae82b	Rev: 1.0	
KiCad E.D.A. 9.0.1			Id: 10/13	



Input filter:
CTRL PWM frequency = 10 kHz
Settling time (10->90%) = ~4 ms
Peak to peak (Duty = 50%) = 0.07 V
Default max current = 3.3 A (with 0.1R Sense resistor)

$$R_{SEN} = V_{REF} / (10 \times I_{TRIP})$$

Current limit = 3.30 A



Designed for: TrendBit
Designed by: Petr.Malaník

Sheet: /System/Peristaltic pump 4/
File: motor.kicad_sch

Title: Peristaltic pump 4

Project: SMPBR

Board: Pump_board

Size: A4

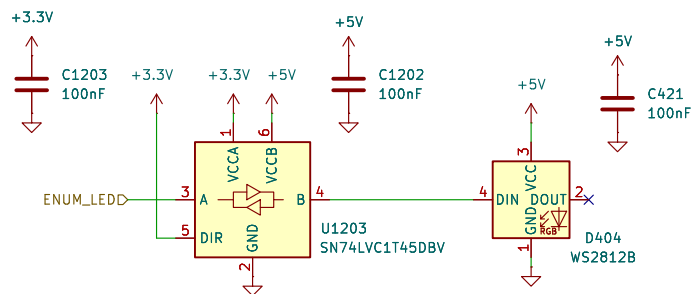
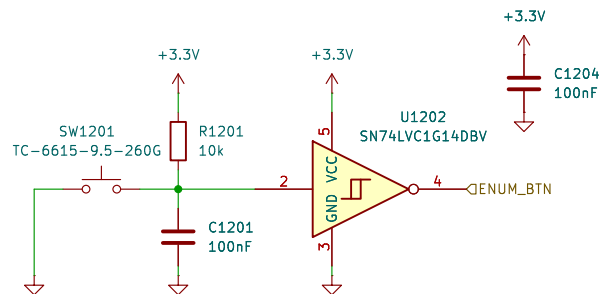
Date: 2024-12-03

GIT hash: 48ae82b

Rev: 1.0

KiCad E.D.A. 9.0.1

Id: 11/13



Designed for: TrendBit
Designed by: Petr_Malaník

Sheet: /System/Enumeration/
File: enumeration.kicad_sch

Title: Enumeration

Project: SMPBR

Board: Pump_board

Size: A4

Date: 2024-12-03

GIT hash: 48ae82b

Rev: 1.0

KiCad E.D.A. 9.0.1

Id: 12/13

